SET NO: 21

Energy Consumption Data Analysis

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Course: DSA06012 Data Handling and Visualization

Question 1

Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

AIM

To import the energy dataset in R and compare the consumption pattern using a histogram and density plot of Units_Consumed.

PROCEDURE

1. Import Energy_Consumption_Data.csv using read.csv() and check it with str() and summary().
2. Plot geom_histogram() on the density scale.
3. Overlay geom_density() and label the plot.

CODE

```
# Import the dataset
library(ggplot2); library(dplyr)
energy <- read.csv("Energy_Consumption_Data.csv")
energy$Month <- factor(energy$Month, levels = c("Jan", "Feb", "Mar"))
str(energy)
summary(energy$Units_Consumed)

# Histogram and density plot
ggplot(energy, aes(x = Units_Consumed)) +
  geom_histogram(aes(y = after_stat(density)), binwidth = 100,
                 fill = "steelblue", colour = "white", alpha = 0.7) +
  geom_density(colour = "red", linewidth = 1) +
  labs(title = "Histogram and Density Plot of Units Consumed",
       x = "Units_Consumed (kWh)", y = "Density")
```

OUTPUT

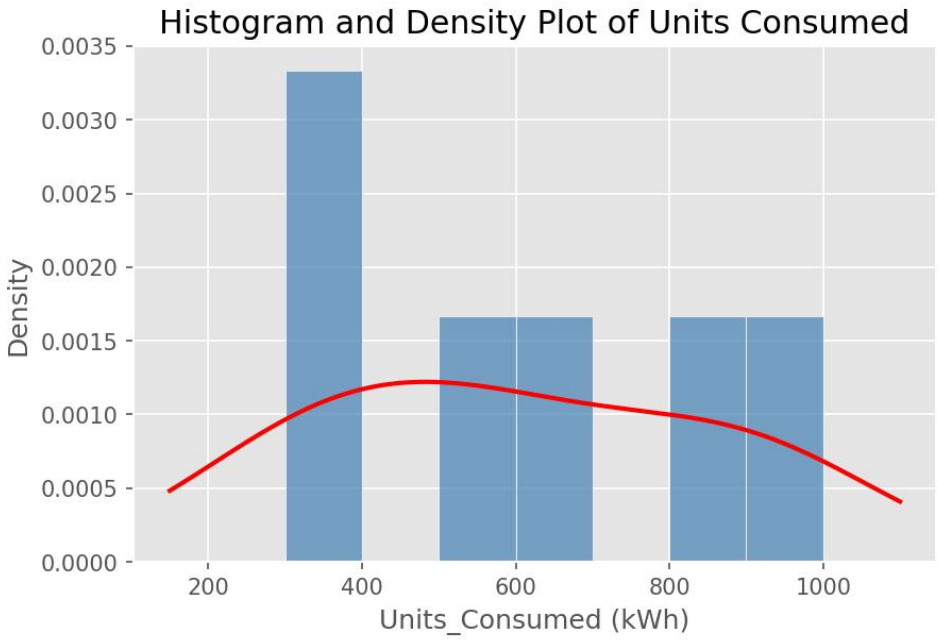


Figure 21.1: Histogram and density plot of Units_Consumed

RESULT

Units_Consumed ranges from 320 to 920 kWh. The consumption is spread over low (320–350 kWh), medium (540–610 kWh) and high (880–920 kWh) levels, with the high level coming from the industrial sector.

Question 2

Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.

AIM

To study the relationship between Temperature and Units_Consumed using a bubble scatter plot with transparency.

PROCEDURE

1. Use geom_point() with Temperature on x and Units_Consumed on y.
2. Map Peak_Hours to size and set alpha = 0.6.
3. Add the title, axis labels and legend.

CODE

```
# Scatter (bubble) plot with transparency
ggplot(energy, aes(x = Temperature, y = Units_Consumed,
                   size = Peak_Hours)) +
  geom_point(alpha = 0.6, colour = "#E15759") +
  scale_size_continuous(range = c(4, 12)) +
  labs(title = "Temperature vs Units Consumed",
       x = "Temperature (°C)", y = "Units_Consumed (kWh)",
       size = "Peak_Hours")
```

OUTPUT

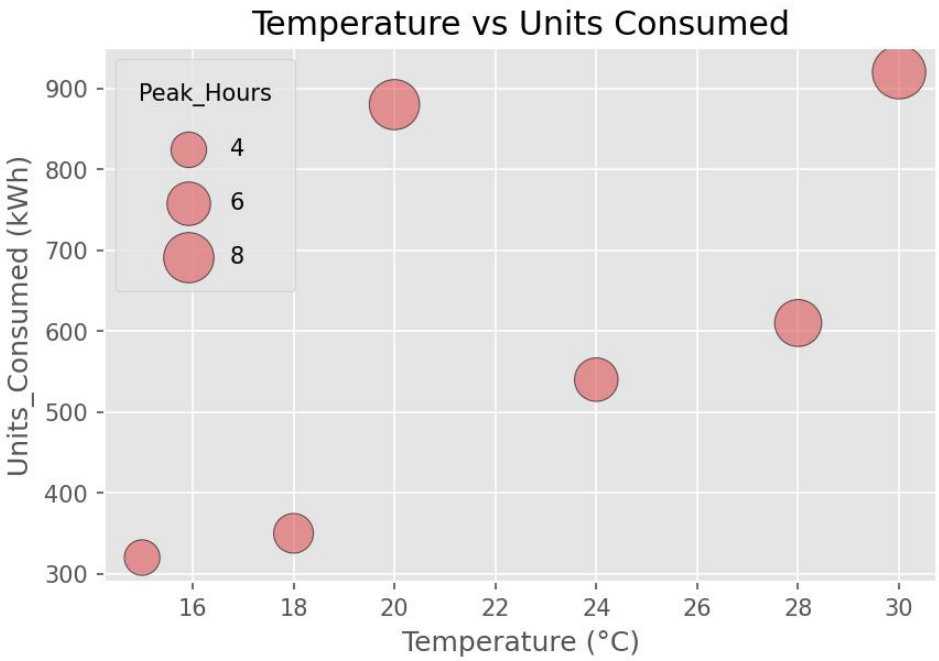


Figure 21.2: Temperature vs Units_Consumed (bubble size = Peak_Hours)

RESULT

Units_Consumed generally increases as Temperature rises, and the larger bubbles (more peak hours) appear at the higher consumption points.

Question 3

Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

AIM

To calculate the average Renewable_Usage of each sector and show it in a bar chart.

PROCEDURE

1. Group the data by Sector using group_by().
2. Calculate the mean Renewable_Usage using summarise().
3. Plot the result with geom_col() and add labels.

CODE

```
# Average Renewable_Usage for each Sector
avg_renew <- energy %>%
  group_by(Sector) %>%
  summarise(Avg_Renewable = mean(Renewable_Usage))
print(avg_renew)

ggplot(avg_renew, aes(x = Sector, y = Avg_Renewable, fill = Sector)) +
  geom_col(width = 0.55) +
  geom_text(aes(label = round(Avg_Renewable, 1)), vjust = -0.5) +
  labs(title = "Average Renewable Usage by Sector",
       x = "Sector", y = "Average Renewable_Usage (%)")
```

OUTPUT

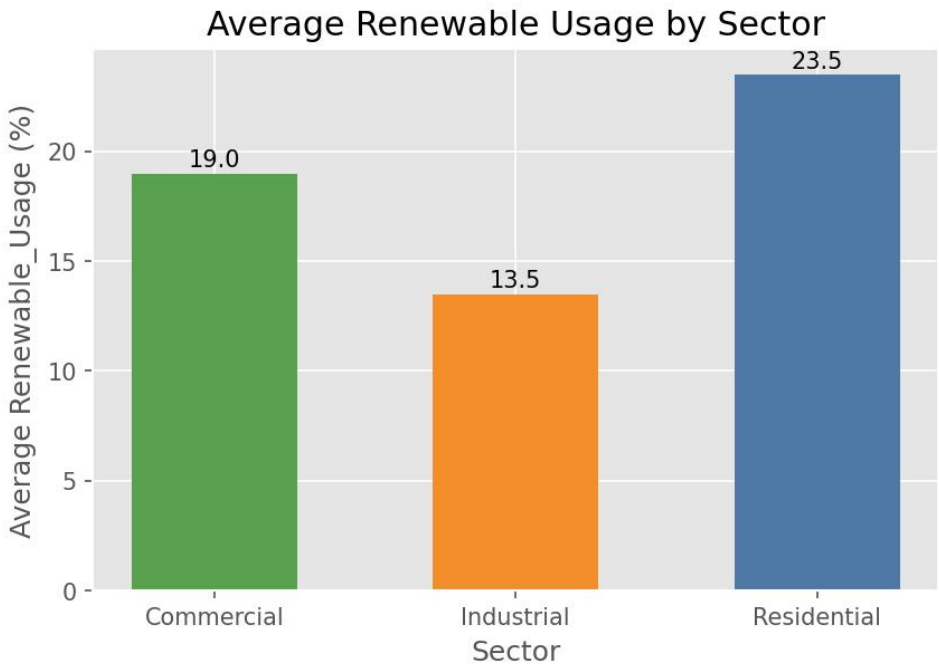


Figure 21.3: Average Renewable_Usage by Sector

RESULT

The average Renewable_Usage is Residential = 23.5%, Commercial = 19.0% and Industrial = 13.5%. Renewable energy adoption is highest in the residential sector and lowest in the industrial sector.

Question 4

Import Energy_Consumption_Data.csv. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.

AIM

To create a monthly usage line chart and a Sector–Region heatmap, and to combine them in an interactive dashboard with Month and Sector filters using R.

PROCEDURE

1. Sum Units_Consumed by Month and plot the line chart with geom_line().
2. Sum Units by Sector and Region and plot the heatmap with geom_tile().
3. Build a shiny dashboard with Month and Sector filters (selectInput) and run it.

CODE

```
library(shiny)

# Line chart: monthly energy usage trend
monthly <- energy %>% group_by(Month) %>%
  summarise(Units = sum(Units_Consumed))
ggplot(monthly, aes(x = Month, y = Units, group = 1)) +
  geom_line(colour = "steelblue", linewidth = 1) + geom_point(size = 3) +
  labs(title = "Monthly Energy Usage Trend",
       x = "Month", y = "Sum of Units_Consumed (kWh)")
```

```
# Heatmap: Sector vs Region vs Units
heat <- energy %>% group_by(Sector, Region) %>%
  summarise(Units = sum(Units_Consumed), .groups = "drop")
ggplot(heat, aes(x = Region, y = Sector, fill = Units)) +
  geom_tile(colour = "white") + geom_text(aes(label = Units)) +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  labs(title = "Heatmap: Sector vs Region (Units_Consumed)")

# Interactive dashboard with Month and Sector filters
ui <- fluidPage(
  titlePanel("Energy Consumption Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput("month", "Month", c("All", levels(energy$Month))),
      selectInput("sector", "Sector", c("All", unique(energy$Sector)))),
    mainPanel(plotOutput("line"), plotOutput("heat"))))

server <- function(input, output) {
  filt <- reactive({
    d <- energy
    if (input$month != "All")  d <- d[d$Month == input$month, ]
    if (input$sector != "All") d <- d[d$Sector == input$sector, ]
    d })
  output$line <- renderPlot({
    d <- filt() %>% group_by(Month) %>% summarise(Units = sum(Units_Consumed))
    ggplot(d, aes(Month, Units, group = 1)) + geom_line() + geom_point() +
      labs(title = "Monthly Energy Usage Trend") })
  output$heat <- renderPlot({
    d <- filt() %>% group_by(Sector, Region) %>%
      summarise(Units = sum(Units_Consumed), .groups = "drop")
    ggplot(d, aes(Region, Sector, fill = Units)) + geom_tile() +
      geom_text(aes(label = Units)) +
      labs(title = "Heatmap: Sector vs Region") })
}
shinyApp(ui, server)
```

OUTPUT

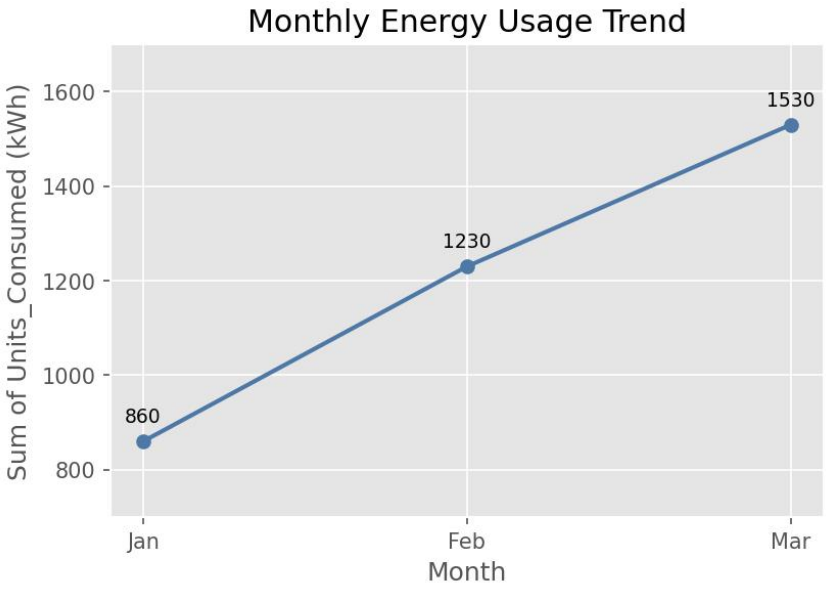


Figure 21.4: Monthly energy usage trend

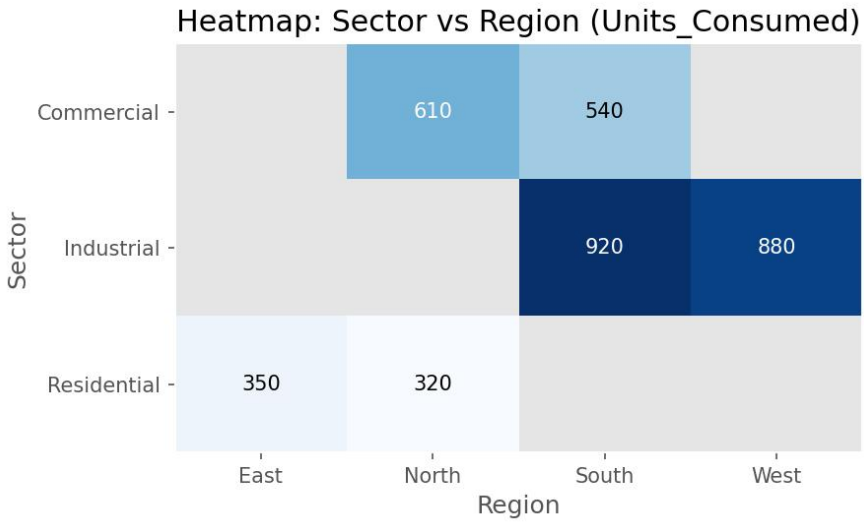


Figure 21.5: Heatmap of Sector vs Region (Units_Consumed)

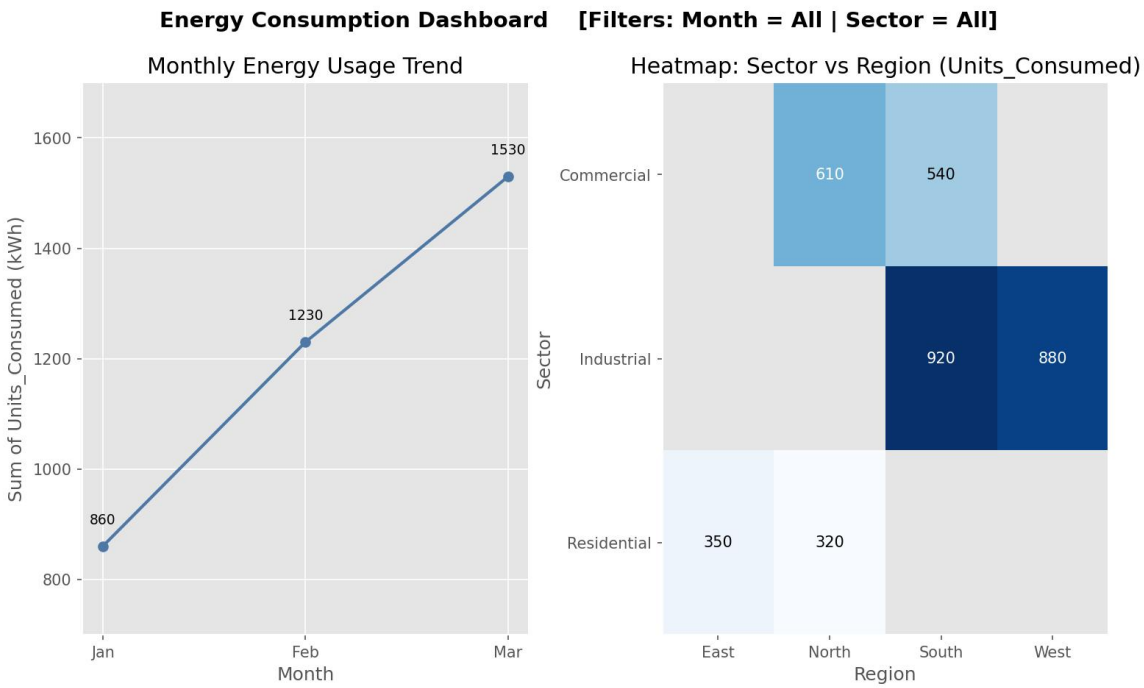


Figure 21.6: Interactive energy consumption dashboard

RESULT

Monthly consumption increases from January (860 kWh) to March (1,530 kWh). The heatmap shows the Industrial sector (West and South) as the highest consumer. The Month and Sector filters update both charts for interactive exploration.